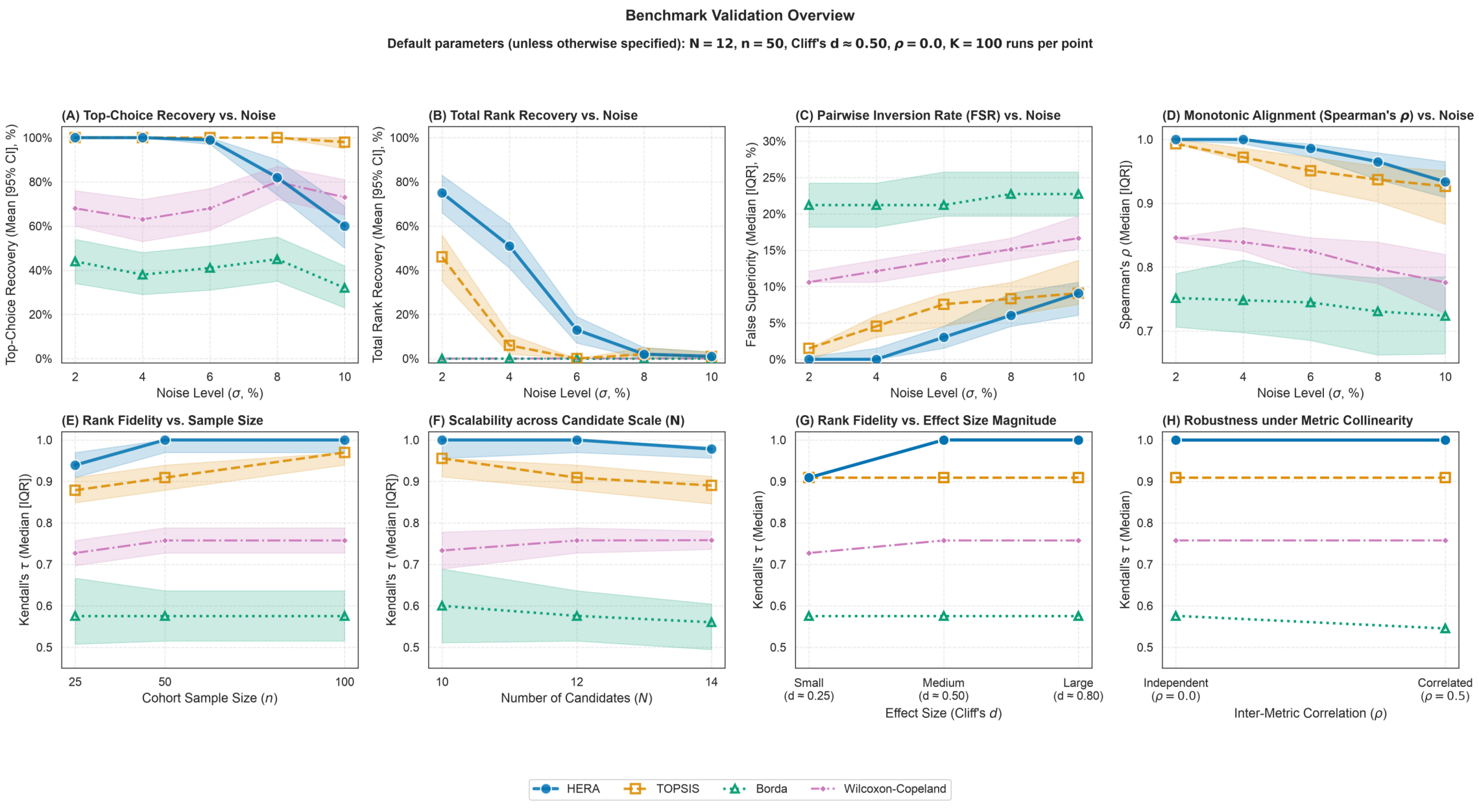
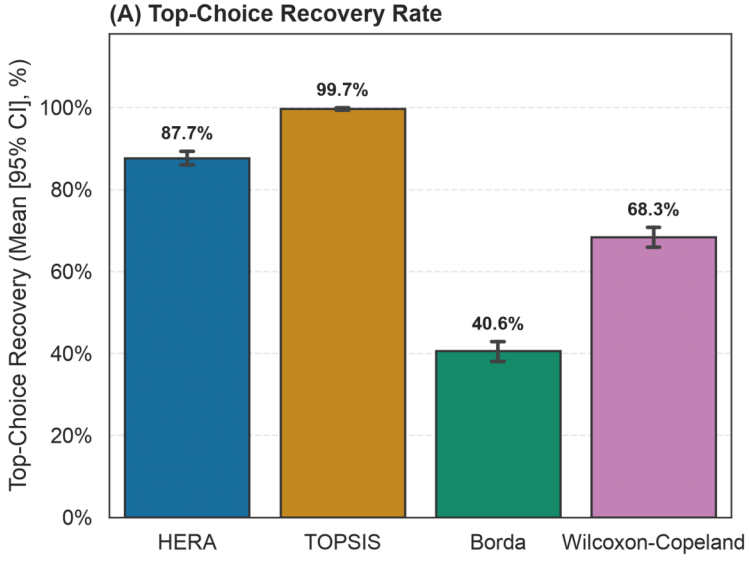
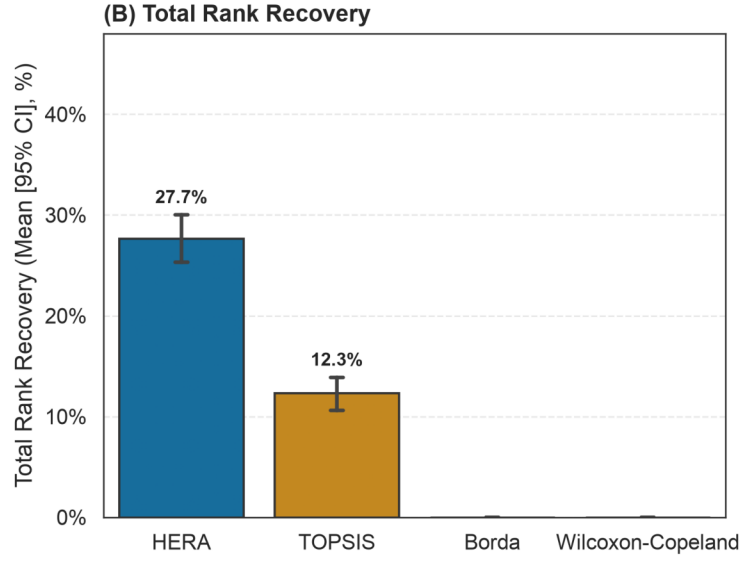
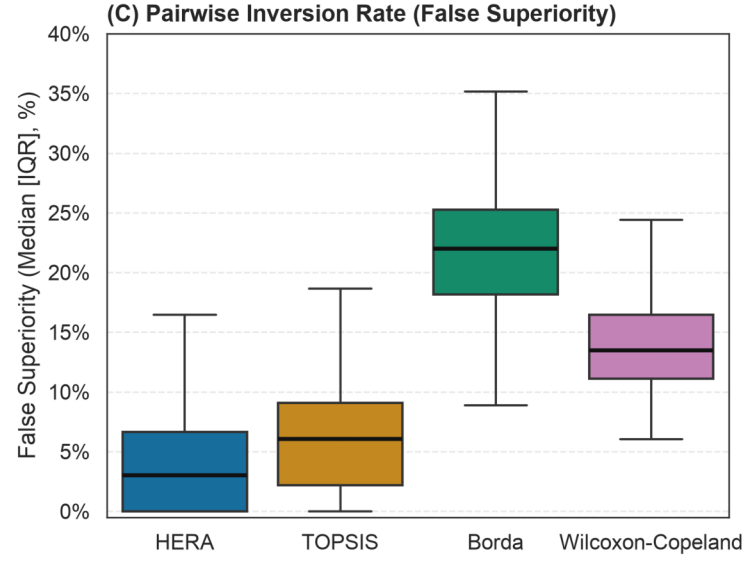

Benchmark Validation Overview
Default parameters (unless otherwise specified): N = 12, n = 50, Cliff's d ≈ 0.50, ρ = 0.0, K = 100 runs per point
(A) Top-Choice Recovery vs. Noise
Top-Choice Recovery (Mean [95% CI], %)
Noise Level (σ, %)
(B) Total Rank Recovery vs. Noise
Total Rank Recovery (Mean [95% CI], %)
Noise Level (σ, %)
(C) Pairwise Inversion Rate (FSR) vs. Noise
False Superiority (Median [IQR], %)
Noise Level (σ, %)
(D) Monotonic Alignment (Spearman's ρ) vs. Noise
Spearman's ρ (Median [IQR])
Noise Level (σ, %)
(E) Rank Fidelity vs. Sample Size
Kendall's τ (Median [IQR])
Cohort Sample Size (n)
(F) Scalability across Candidate Scale (N)
Kendall's τ (Median [IQR])
Number of Candidates (N)
(G) Rank Fidelity vs. Effect Size Magnitude
Kendall's τ (Median)
Small (d ≈ 0.25)
Medium (d ≈ 0.50)
Large (d ≈ 0.80)
Effect Size (Cliff's d)
(H) Robustness under Metric Collinearity
Kendall's τ (Median)
Independent (ρ = 0.0)
Correlated (ρ = 0.5)
Inter-Metric Correlation (ρ)
HERA
TOPSIS
Borda
Wilcoxon-Copeland

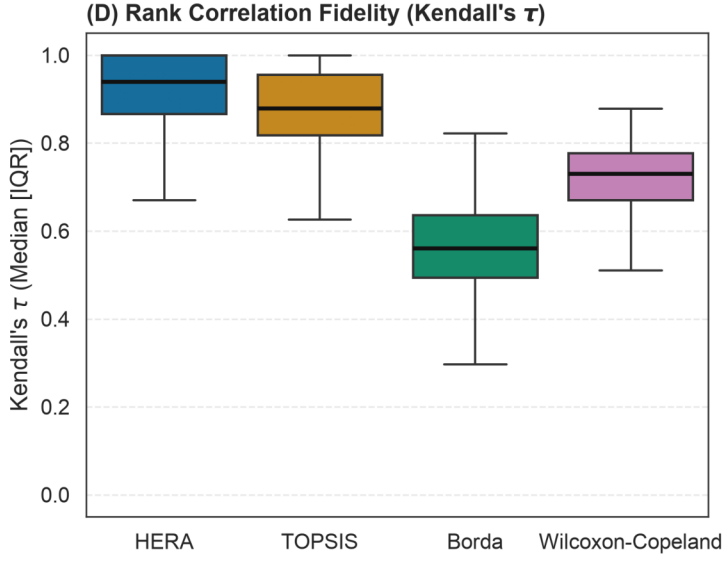
Pooled Distribution across Core Conditions

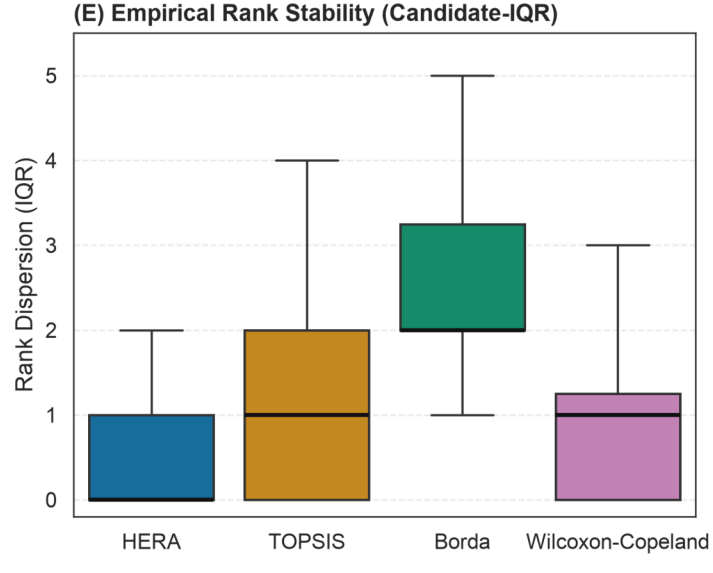
N ∈ {10, 12, 14}, σ ∈ {2%, 4%, 6%, 8%, 10%}, n = 50, ρ = 0.0; K = 1500 runs per method

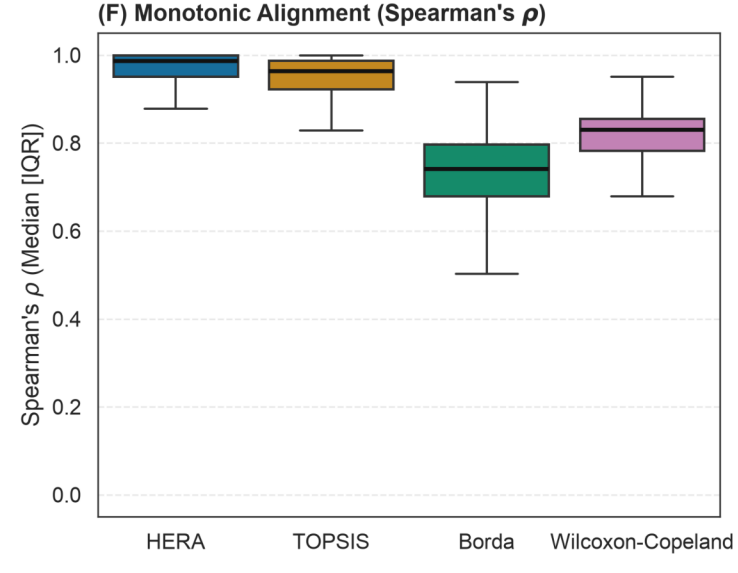


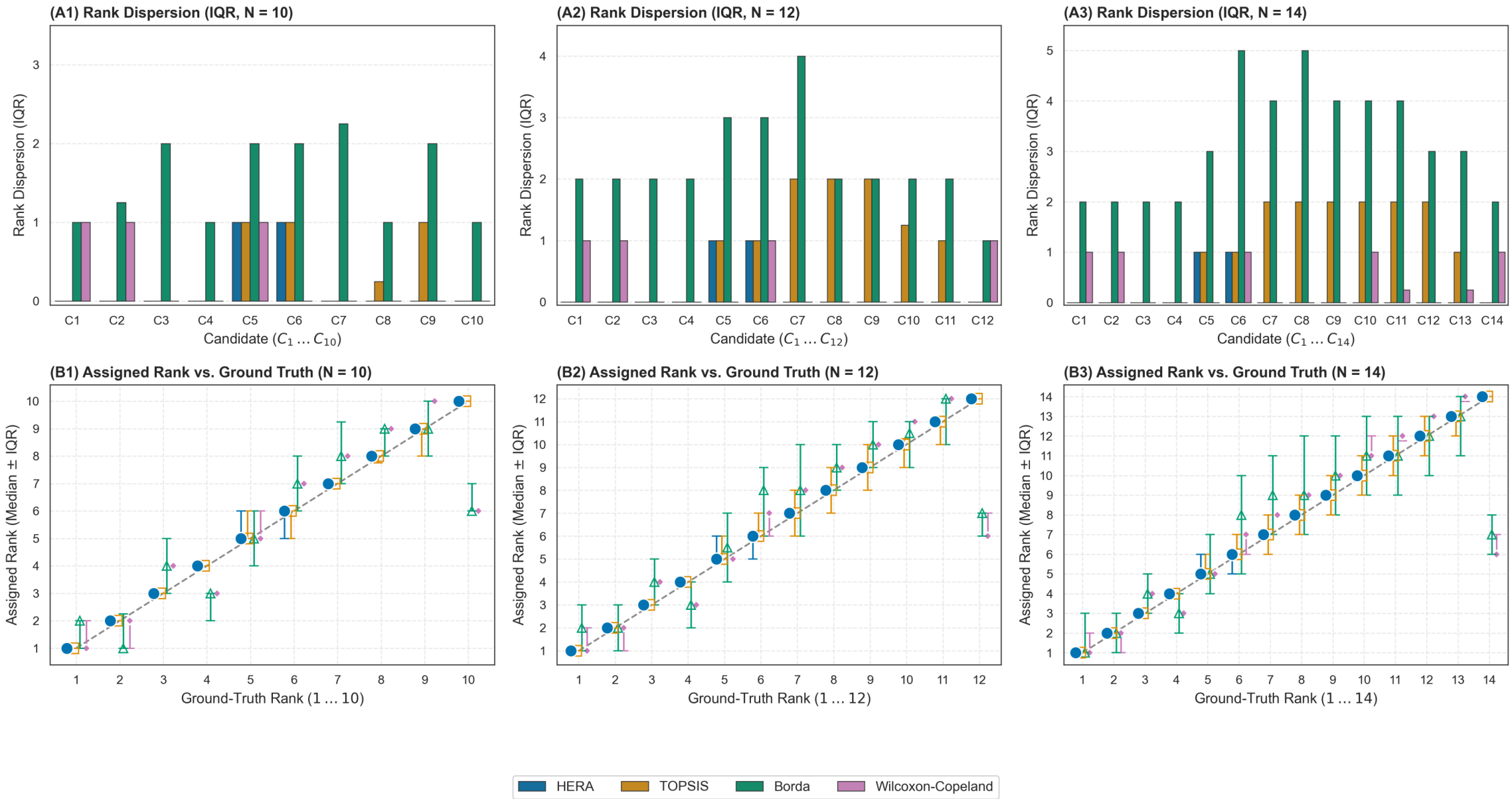


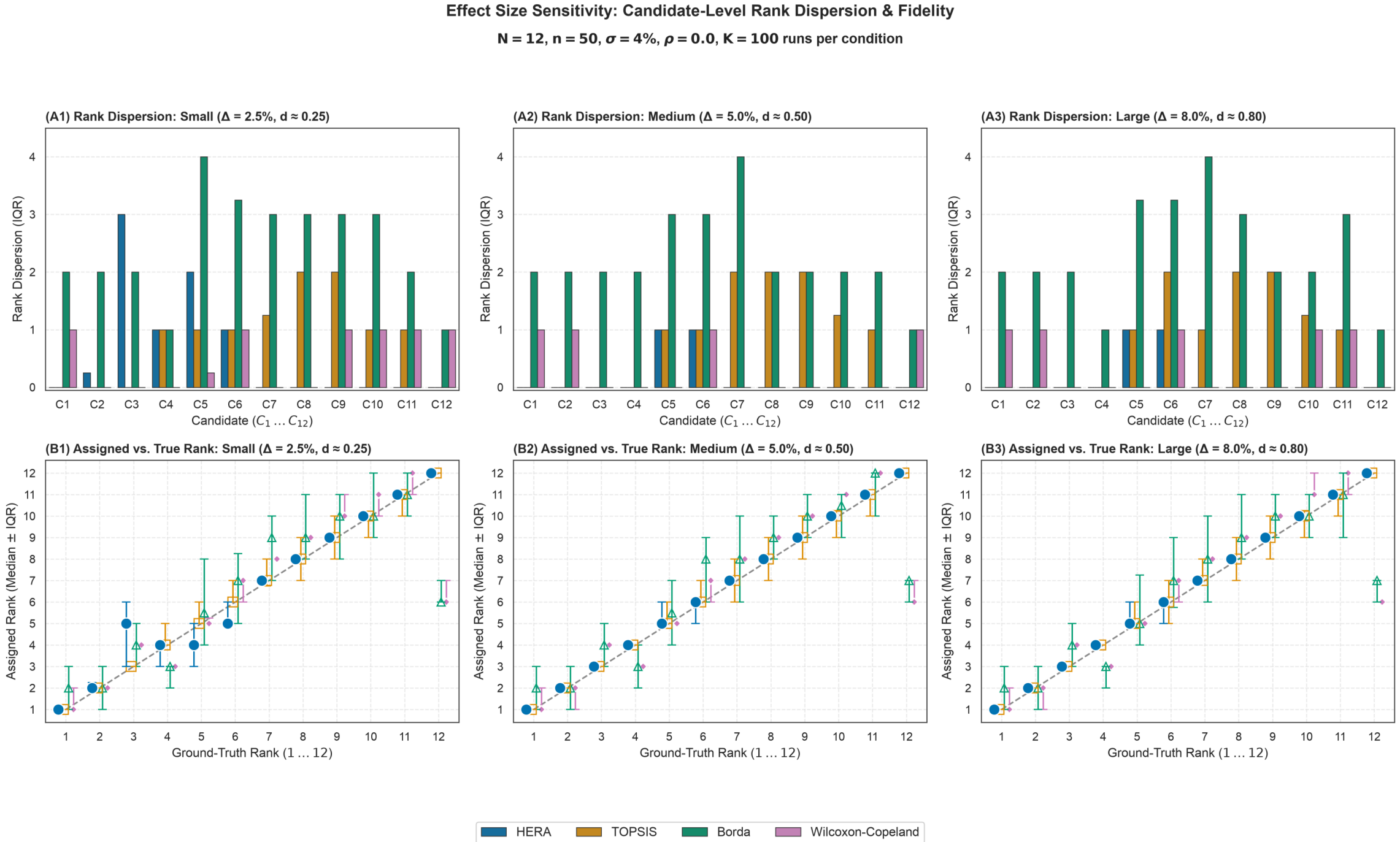






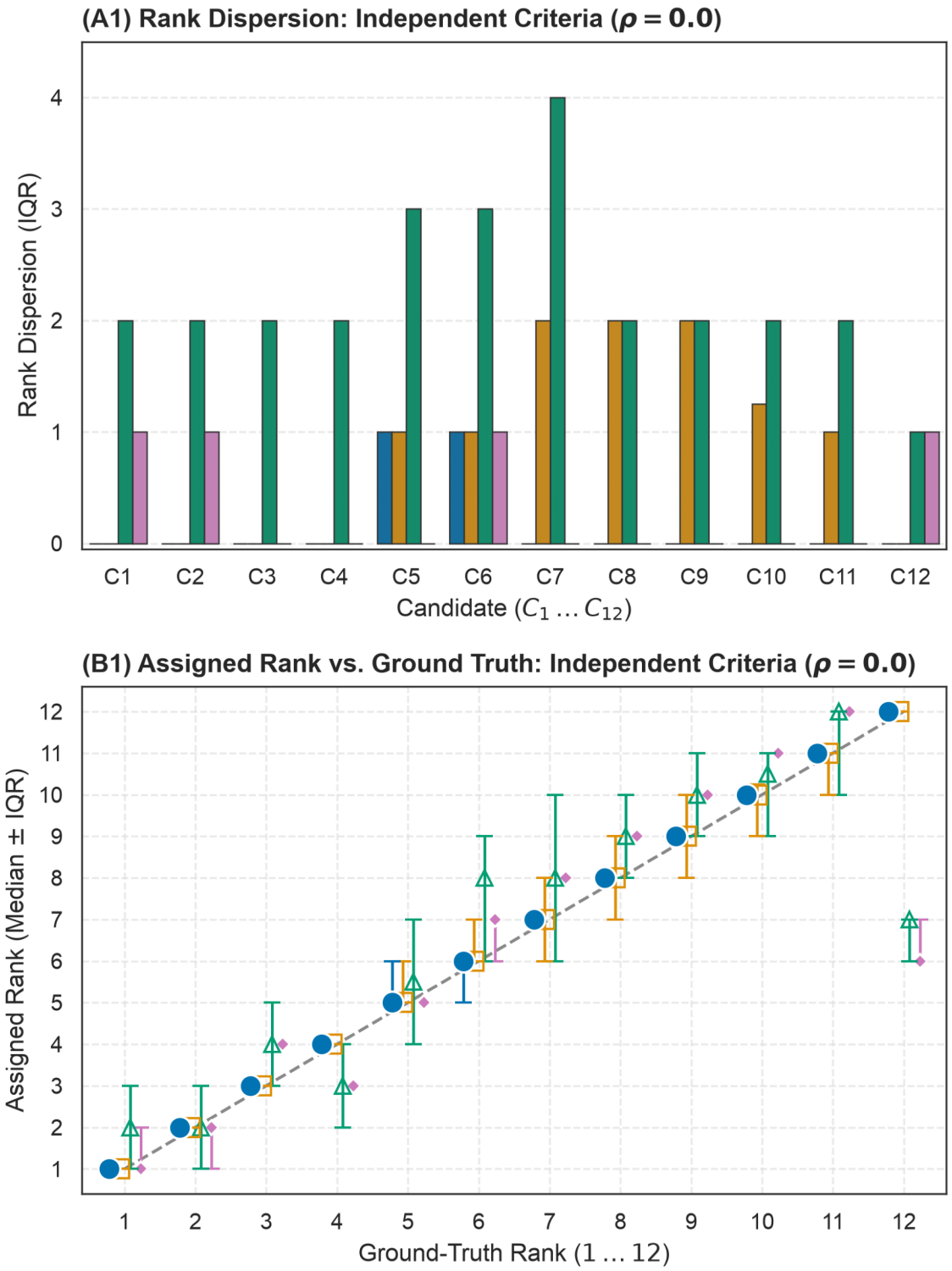


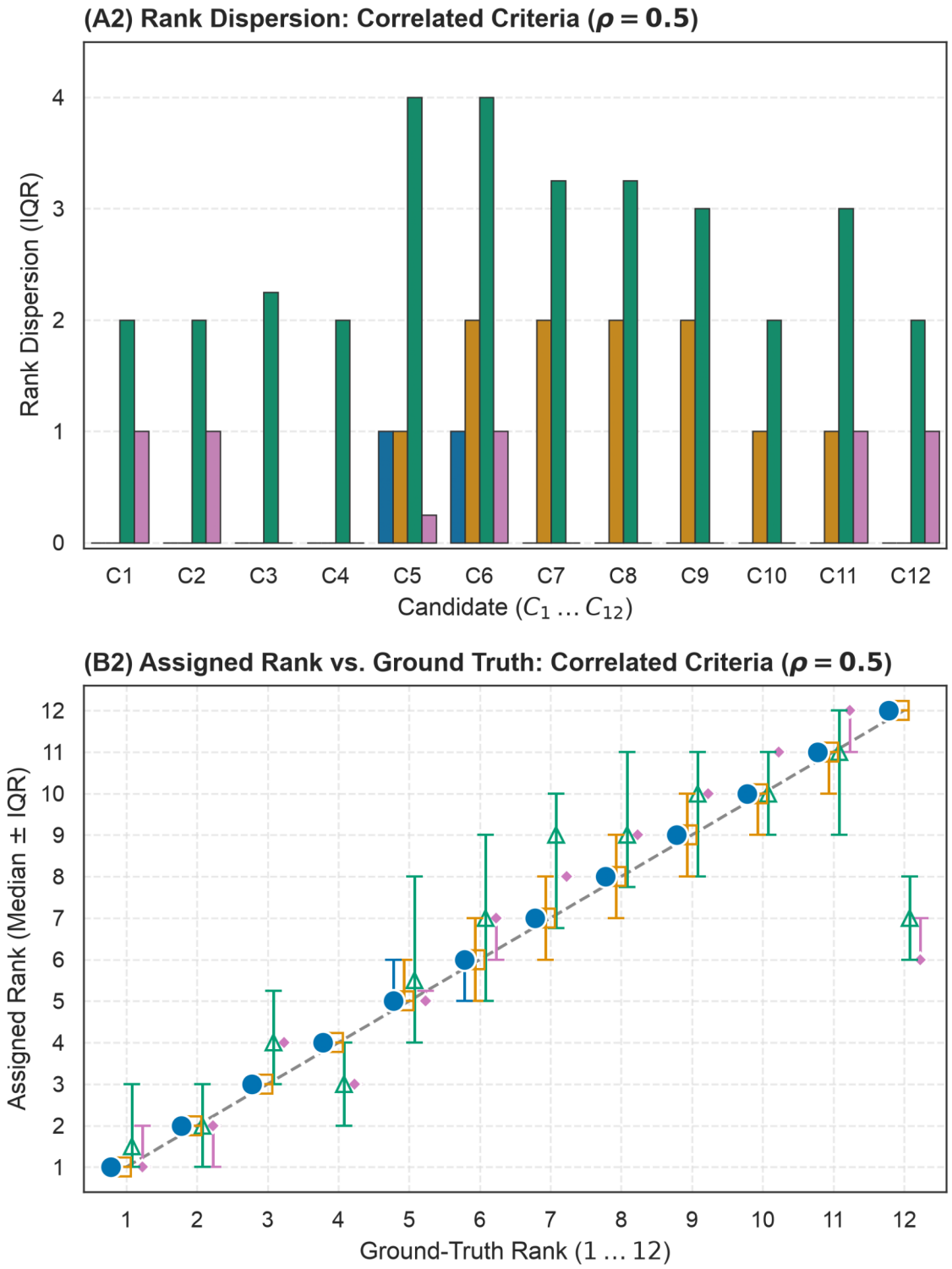

Candidate-Level Rank Stability Across Scales
N ∈ {10, 12, 14}, n = 50, σ = 4%, ρ = 0.0, K = 100 runs per point
(A1) Rank Dispersion (IQR, N = 10)
Rank Dispersion (IQR)
Candidate (C_1 … C_{10})
(A2) Rank Dispersion (IQR, N = 12)
Rank Dispersion (IQR)
Candidate (C_1 … C_{12})
(A3) Rank Dispersion (IQR, N = 14)
Rank Dispersion (IQR)
Candidate (C_1 … C_{14})
(B1) Assigned Rank vs. Ground Truth (N = 10)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 10)
(B2) Assigned Rank vs. Ground Truth (N = 12)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 12)
(B3) Assigned Rank vs. Ground Truth (N = 14)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 14)
HERA
TOPSIS
Borda
Wilcoxon-Copeland


Effect Size Sensitivity: Candidate-Level Rank Dispersion & Fidelity
N = 12, n = 50, σ = 4%, ρ = 0.0, K = 100 runs per condition
(A1) Rank Dispersion: Small (Δ = 2.5%, d ≈ 0.25)
(A2) Rank Dispersion: Medium (Δ = 5.0%, d ≈ 0.50)
(A3) Rank Dispersion: Large (Δ = 8.0%, d ≈ 0.80)
Rank Dispersion (IQR)
Candidate ($C_1 \dots C_{12}$)
C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12
(B1) Assigned vs. True Rank: Small (Δ = 2.5%, d ≈ 0.25)
(B2) Assigned vs. True Rank: Medium (Δ = 5.0%, d ≈ 0.50)
(B3) Assigned vs. True Rank: Large (Δ = 8.0%, d ≈ 0.80)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 12)
HERA
TOPSIS
Borda
Wilcoxon-Copeland

Inter-Metric Collinearity Sensitivity: Rank Dispersion & Fidelity

N = 12, n = 50, σ = 4%, Cliff's d ≈ 0.50, K = 100 runs per condition





HERA TOPSIS Borda Wilcoxon-Copeland